

Student Enrollment & Course Management System (SECMS)

6. Project Development Phase:

Project Flow:

Milestone 1: Salesforce Developer Account Creation

Milestone 2: Custom Object Creation (Student, Course, Instructor, Enrollment)

Milestone 3: Fields Creation

Milestone 4: Create Relationships

Milestone 5: Validation Rules

Milestone 6: Approval Process

Milestone 7: Tabs Creation

Milestone 8: Lightning App Creation

Milestone 9: Page Layouts and Record Types

Milestone 10: Flows (Automation)

Milestone 11: Reports

Milestone 12: Dashboards

Milestone 13: Security Setup

Milestone 14: Conclusion

What we learned

1. Implementing a real-time Salesforce project
2. Designing custom objects and relationships
3. Creating page layouts and record types
4. Writing validation rules
5. Building automation with flows
6. Creating approval workflows
7. Developing reports and dashboards
8. Setting up security with profiles, permission sets and sharing
9. Testing the application end to end

Milestone 1 – Salesforce Developer Account:

Introduction:

Salesforce is a cloud-based customer success platform that lets organizations manage relationships, automate business processes and store data securely online. Before building SECMS, our team first got familiar with the platform and its basic features.

Because Salesforce runs entirely in the cloud, we could design, configure and test the whole enrollment system without installing any software.

What is Salesforce?

Salesforce is a platform for sales, service, marketing, analytics and customer engagement. It lets a business manage contacts, follow-ups and records in one place instead of scattered files.

For our project we used the platform's point-and-click tools (objects, flows, approvals, reports and dashboards) to build a complete academic enrollment solution. The video below gives a quick introduction to Salesforce.

<https://youtu.be/r9EX3lGde5k>

Activity 1: Creating Developer Account:

We created a developer org in Salesforce as shown below.

Go to <https://developer.salesforce.com/signup>

On the sign-up form we entered the following details:

First name & Last name

Email

Role : Developer

Company : College Name

Country : India

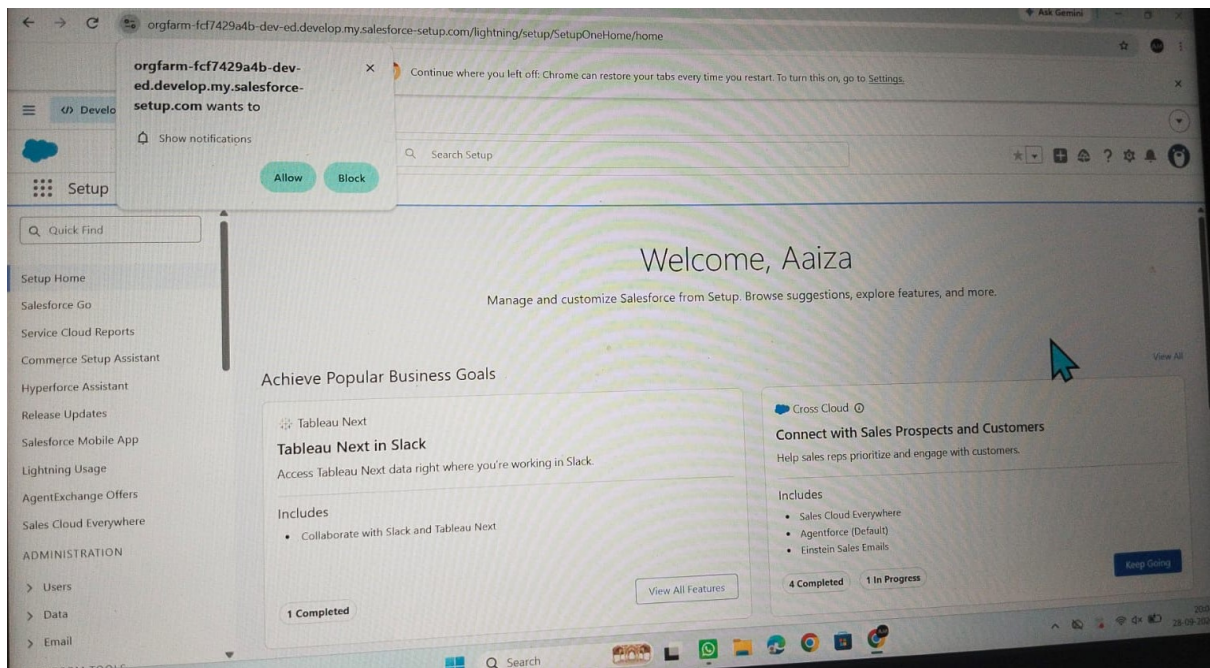
Postal Code : Postal (PIN) code

Username : a combination of our name and the college name

The username does not need to be a real email address; we used the format username@organization.com and clicked Sign me up. After submitting the form, we waited for the verification email.

Activity 2: Account Activation:

1. Open the inbox of the email address used at sign-up and click Verify Account (the email can take 5–10 minutes to arrive).
2. Click on Verify Account
3. Set a password, answer the security question and click Change Password.
4. After this we were redirected to the Salesforce Setup page.



Milestone 2 – Objects:

Activity 1: Creating the Student Object:

We created the Student custom object to store student information such as personal details, contact information and enrollment status.

Steps to create the Student object:

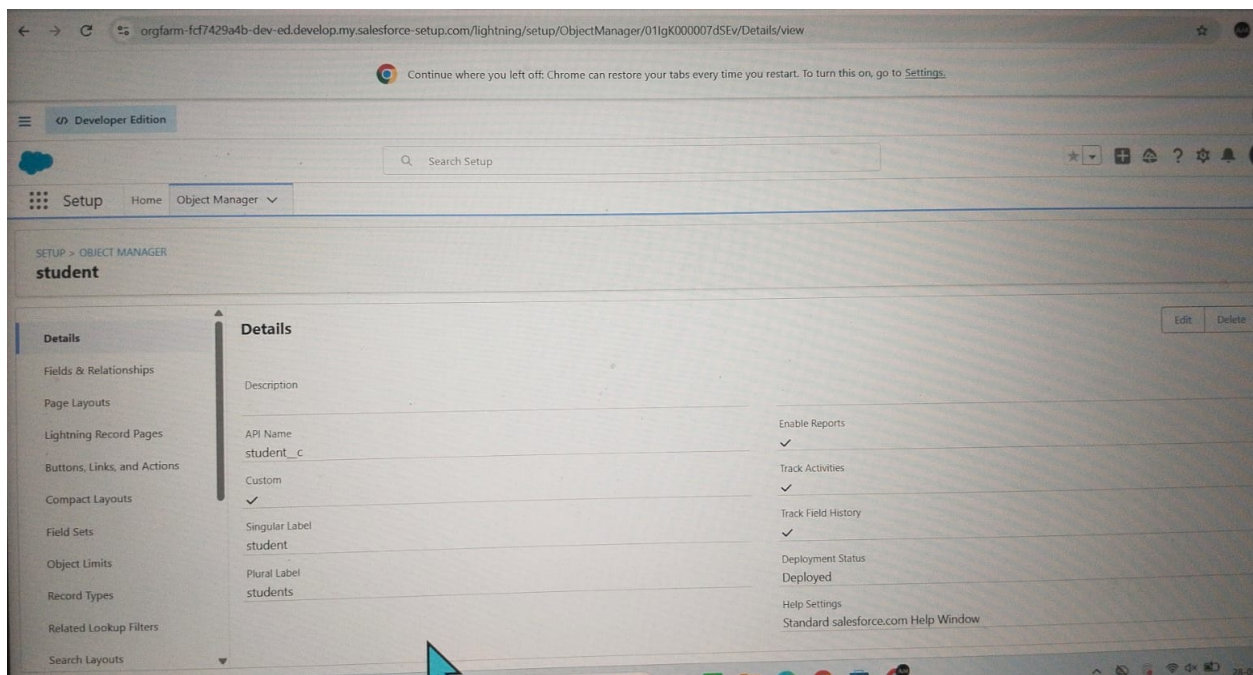
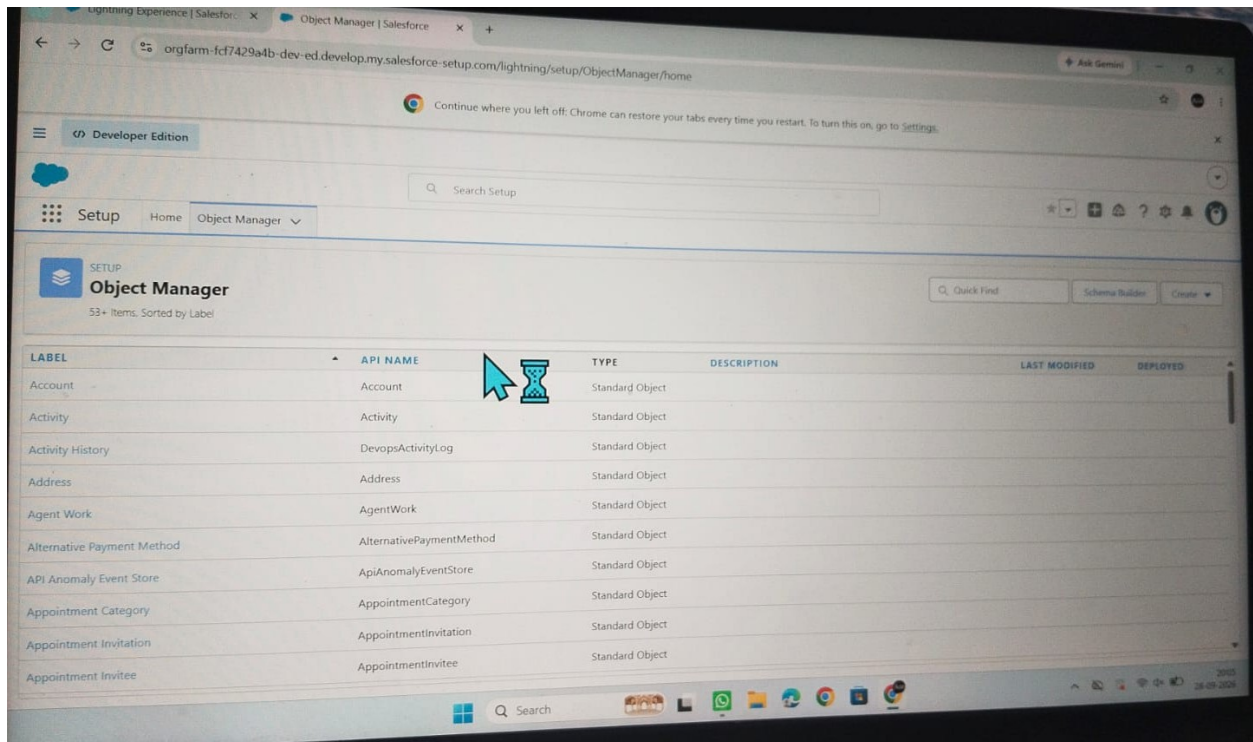
Go to the Setup page.

Click on Object Manager.

Click on Create → Custom Object.

Enter the details as below:

1. Label: Student
2. Plural Label: Students
3. Object Name: Student
4. Record Name: Student Name
5. Data Type: Text
6. Under Optional Features, select:
7. Allow Reports
8. Allow Activities
9. Track Field History
10. Under Search Status, select:
11. Allow Search
12. Click Save.



In the same way, we created the **Courses**, **Instructors**, and **Enrollment** objects

Milestone 3 - Fields:

Table 1: Student Object Fields:

Field Label	Data Type
Student Name	Text (Standard)
Email	Email
Phone	Phone
Date of Birth	Date
Student Status	Picklist
Address	Text Area

Table 2: Course Object Fields:

Field Label	Data Type
Course Name	Text (Standard)
Category	Picklist
Duration	Number/Text
Course Fee	Currency
Description	Text Area

Table 3: Instructor Object Fields:

Field Label	Data Type
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Instructor Name	Text (Standard)
Instructor Code	Text
Expertise	Picklist
Phone	Phone
Email	Email

Table 4: Enrollment Object Fields:

Field Label	Data Type
Student	Lookup (Student)
Course	Lookup (Course)
Instructor	Lookup (Instructor)
Enrollment Date	Date
Enrollment Status	Picklist
Fees Paid	Checkbox
Total Amount	Currency
Comments	Text Area

Milestone 4 – Creating Relationships:

1. Activity 1: Create Lookup from Enrollment to Student:

1.

Go to Setup → click Object Manager.

2. Type Enrollment in Quick Find → click Enrollment__c object.

3. Click Fields & Relationships → click New.

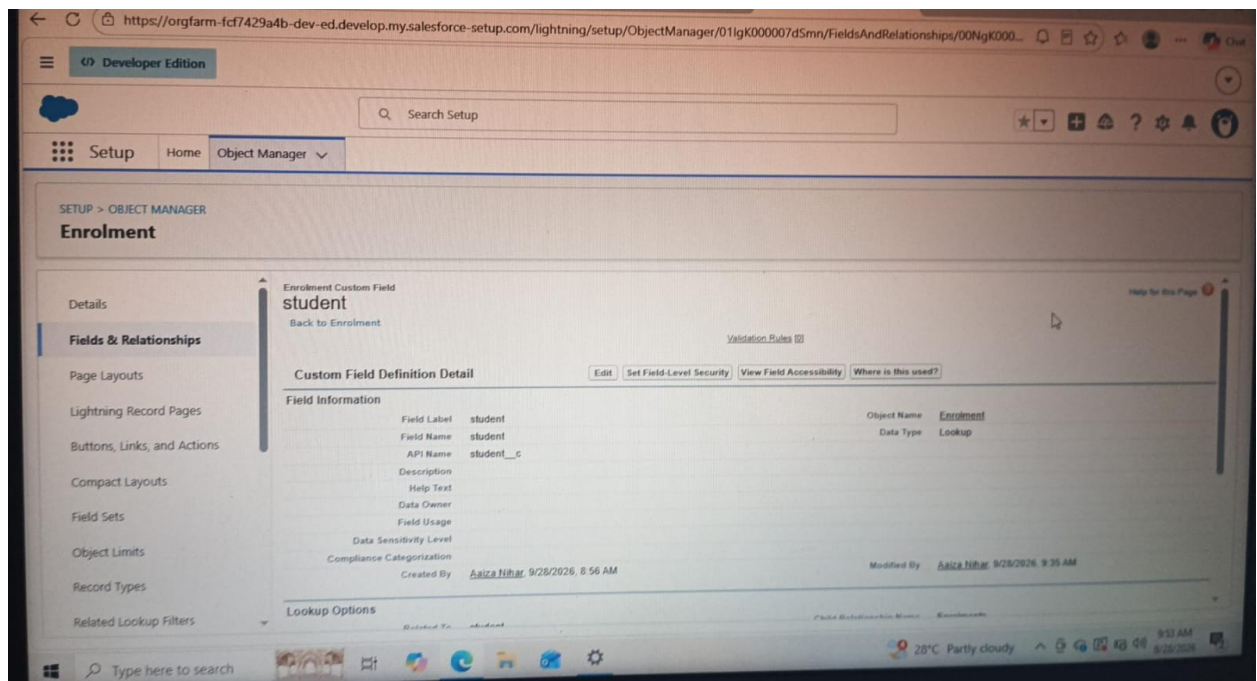
4. Select Lookup Relationship → click Next.

5. For Related To, select Student__c → click Next.

6. Enter Field Label: Student (API Name: Student__c).

7. Click Next through Field-Level Security and Layout options (ensure it's visible and added to layout).

8. Click Save.



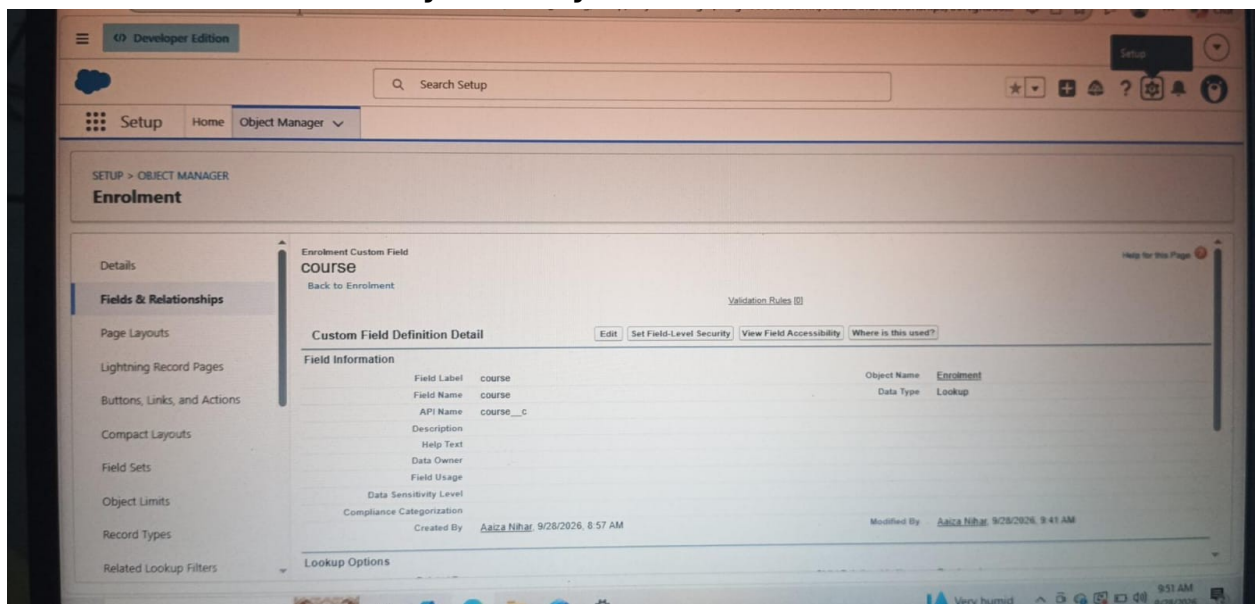
Activity 2: Create Lookup from Enrollment to Course

Purpose:

To link every Enrollment record to a Course.

Steps:

1. Go to Setup → Object Manager → Enrollment__c.
2. Click Fields & Relationships → New.
3. Select Lookup Relationship → Next.
4. Related To: select Course__c → click Next.
5. Field Label: Course (API Name: Course__c).
6. Click Next (set visibility and layout) → Save.



Activity 3: Create Lookup from Enrollment to Instructor:

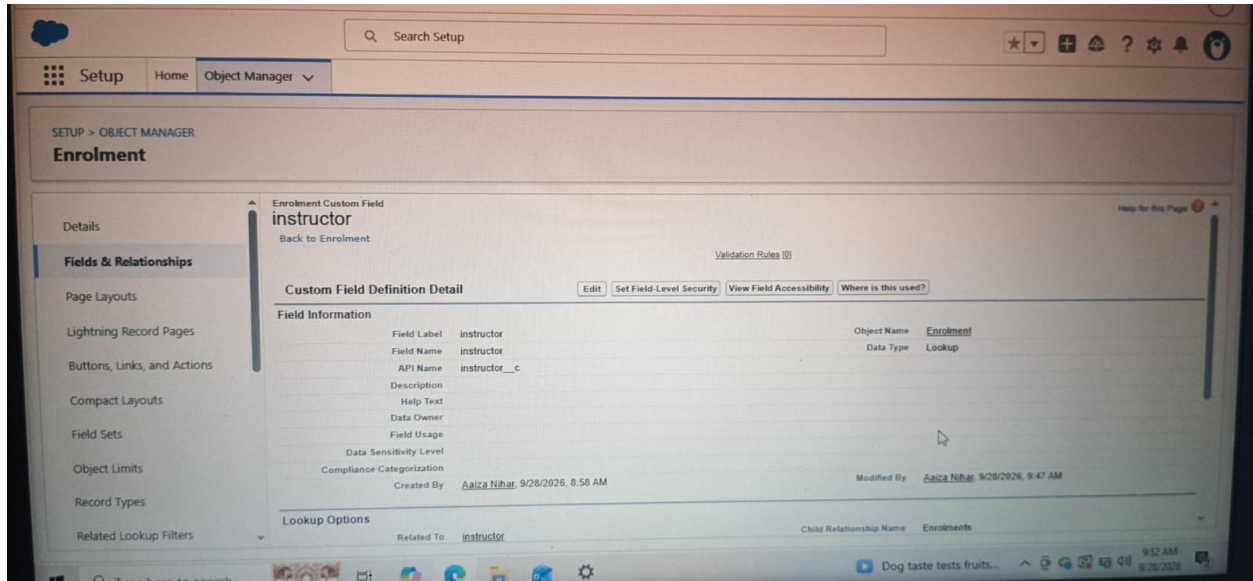
Purpose:

To record which Instructor is assigned to a particular Enrollment.

Steps:

1. Go to Setup → Object Manager → Enrollment__c.
2. Click Fields & Relationships → New.
3. Choose Lookup Relationship → Next.

4. Related To: select Instructor__c → Next.
5. Field Label: Instructor (API Name: Instructor__c).
6. Complete Field-Level Security and Page Layout options → Save



Milestone 5 – Validation Rules:

Activity 1: Student Email Domain Validation

Purpose:

To make sure only official institutional email addresses are accepted for students.

Steps:

Go to Setup → Object Manager → Student__c.

Click Validation Rules → New.

Rule Name: Student_Email_Domain_Validation.

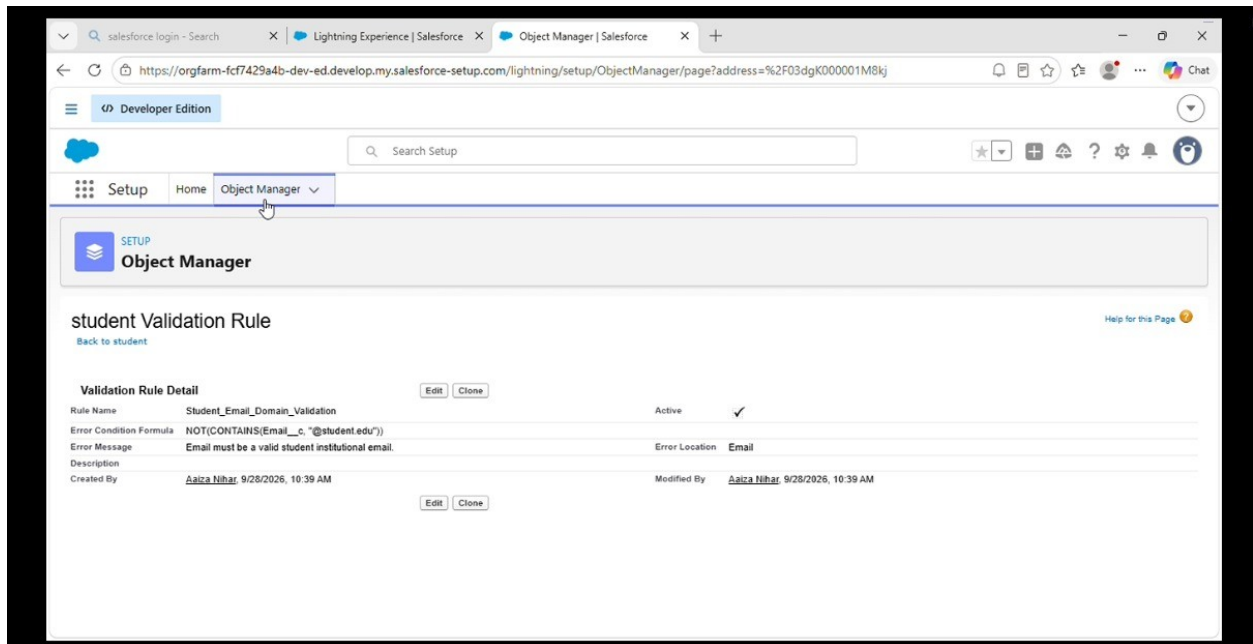
Error Condition Formula: (example for @student.edu)

- NOT(CONTAINS(Email__c, "@student.edu"))

Error Message: Email must be a valid student institutional email.

Error Location: Field → Email__c.

Click Save.



Activity 2: Fees Must Be Paid Before Approval

Purpose:

Prevents an Enrollment from being approved until the fees are paid.

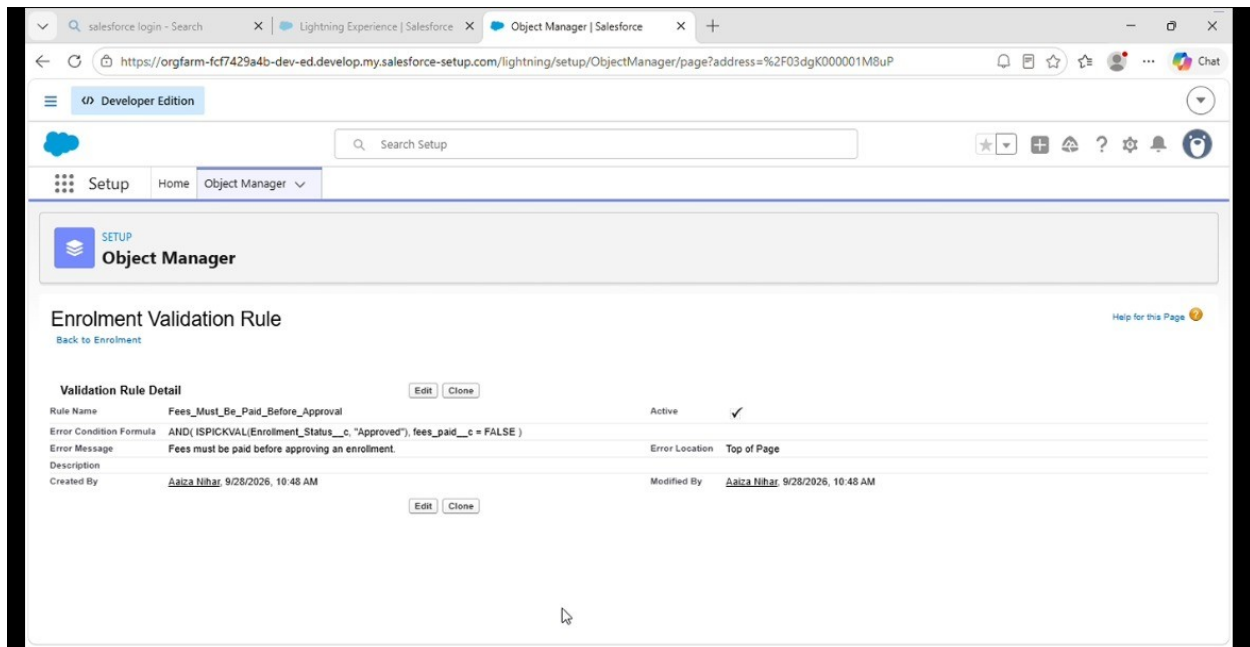
Steps:

1. Go to Setup → Object Manager → Enrollment__c.
2. Click Validation Rules → New.
3. Rule Name: Fees_Must_Be_Paid_Before_Approval.
4. Error Condition Formula:
AND(
 ISPICKVAL(Enrollment_Status__c, "Approved"),
 Fees_Paid__c = FALSE
)

Error Message: Fees must be paid before approving an enrollment.

Error Location: Top of Page (or a field, as required).

Click Save.



Milestone 6 – Approval Process

Activity 1: Create Approval Process for Enrollment Requests

Purpose:

Routes Enrollment requests to the Training Manager for approval or rejection.

Steps:

1. Go to Setup → Approval Processes → select Enrollment__c.
2. Click Create New Approval Process → Use Standard Setup.
3. Process Name: Approval_Enrollment_Request.

4. Entry Criteria: Enrollment_Status__c = "Requested".
5. Save and configure steps.
6. Add Approval Step:
7. Approver: specific Training Manager user.
8. Under Final Approval Actions:
9. Add Field Update to set Enrollment_Status__c = "Approved".
10. Under Final Rejection Actions:
11. Add Field Update to set Enrollment_Status__c = "Rejected".
12. Add Email Alert to send rejection email to student.
13. Activate the approval process.

Milestone 7- Tabs

Activity 1: Creating a Tab for the Student Object:

Go to Setup.

1. In the Quick Find box, type Tabs.
2. Under Custom Object Tabs, click New.
3. Select Student from the object list.
4. Choose a tab style (icon).
5. Click Next, assign it to all profiles, and then click Save.

Activity 2: Creating Remaining Tabs

Then create tabs for the remaining objects: Courses, Instructors and Enrollment. Follow the same steps as in Activity 1.

Milestone 8 - The Lightning App

Activity 1: Creating the Lightning Application for the Project

Navigate to Setup.

Enter App Manager in the Quick Find search box.

Click New Lightning App.

Provide the app details:

App Name: Student Enrollment & Course Management

Developer Name auto-fills

Description: A short description of the purpose of the app (optional)

Click Next and choose the following settings:

App Branding (optional): Add a logo or choose a colour theme

Navigation Style: Choose Standard Navigation

Click Next and assign the app to specific profiles (e.g., **Admin, Enrollment Officer, Instructor**).

Click Next to select the items (tabs) to be included in the app. Add:

1. Student
2. Course
3. Instructor
4. Enrollment
5. Reports
6. Dashboards
7. Home
8. Tasks

Click Next and review the app settings.

Finally, click Save & Finish.

Milestone 9 – Page Layouts and Record Types:

Activity 1: Create Record Types for Enrollment:

Purpose:

To separate the New Enrollment and Re-Enrollment processes.

Steps:

1. Go to Setup → Object Manager → Enrollment__c.

2. Click Record Types → New.

3. For the first record type:

- Existing Record Type: Master
- Record Type Label: New Enrollment
- Enable for required profiles (Admin, Enrollment Officer).
- Select the default Page Layout → Save & New.

4. For the second record type:

- Record Type Label: Re-Enrollment
- Enable for the same profiles.
- Select the default Page Layout → Save.

Activity 2: Creating Separate Page Layouts for Each Enrollment Record Type

Purpose:

To display different fields for New Enrollment and Re-Enrollment records.

Steps:

1. Enrollment__c object → click Page Layouts.

2. Click New to create a layout for New Enrollment:

- Clone from existing Enrollment Layout → name it New Enrollment Layout → Save.
- Keep fields relevant for first-time enrollment: Student, Course, Instructor, Enrollment Date, Status, Fees, etc.
- Remove unnecessary re-enrollment fields if any.
- Click Save.

3. Again, click New to create Re-Enrollment Layout:

- Clone existing layout → name it Re-Enrollment Layout → Save.
- Include fields like previous Enrollment reference (if any), Re-Enrollment Reason, Status, Fees, etc.
- Click Save.

4. Click Page Layout Assignment → Edit Assignment.

5. For each profile and record type:

- New Enrollment → use New Enrollment Layout.

Re-Enrollment → use Re-Enrollment Layout.

Click Save.

Activity 3: Adding Enrollment Related Lists to Parent Objects:

Purpose:

To view all Enrollments directly from the Student, Course and Instructor records.

a) Student → Enrollment Related List

Go to Setup → Object Manager → Student__c.

Click Page Layouts → open the main layout.

In the palette, select Related Lists.

Drag Enrollments related list onto the layout.

Customize columns if needed → Save.

b) Course → Enrollment Related List

Go to Object Manager → Course__c → Page Layouts.

Open layout → add Enrollments related list.

Save.

c) Instructor → Enrollment Related List

Go to Object Manager → Instructor__c → Page Layouts.

Open layout → add Enrollments related list.

Save.

Milestone 10 – Automation with Flow

Activity 1: Auto-Populate Enrollment Date (Record-Triggered Flow)

Purpose:

Sets Enrollment_Date__c to today's date when an Enrollment is created.

Steps:

1. Go to Setup → Flows → New Flow.
2. Select Record-Triggered Flow → Create.
3. Object: Enrollment__c.
4. Trigger: A record is created.
5. Optimize for Actions and Related Records → Done.
6. Add Update Records element:
 - Label: Set Enrollment Date.
 - Use a record that triggered the flow.
 - Field: Enrollment_Date__c → Value: \$Flow.CurrentDate.
7. Save as SetEnrollmentDate → Activate.

Activity 2: Send Email When Enrollment Approved

Purpose:

When Enrollment_Status__c = Approved, the flow sends an email to the student, alerts the admin and sets the Student status to Active.

Steps:

1. Create Email Template and Email Alert for student notification.
2. Go to Flows → New → Record-Triggered Flow.
3. Object: Enrollment__c, Trigger on update.
4. Condition: Enrollment_Status__c Equals "Approved".
5. Add Get Records to fetch related Student__c.
6. Add Action (Email Alert) to send email to student.
7. Add Action to notify Admin if needed.
8. Add Update Records to set Student__c.Status__c = "Active".
9. Save as Enrollment Approved Action → Activate.

Activity 3: Auto-Assign Instructor Based on Course Category

Purpose:

Assigns an Instructor automatically based on the Course category.

Rules:

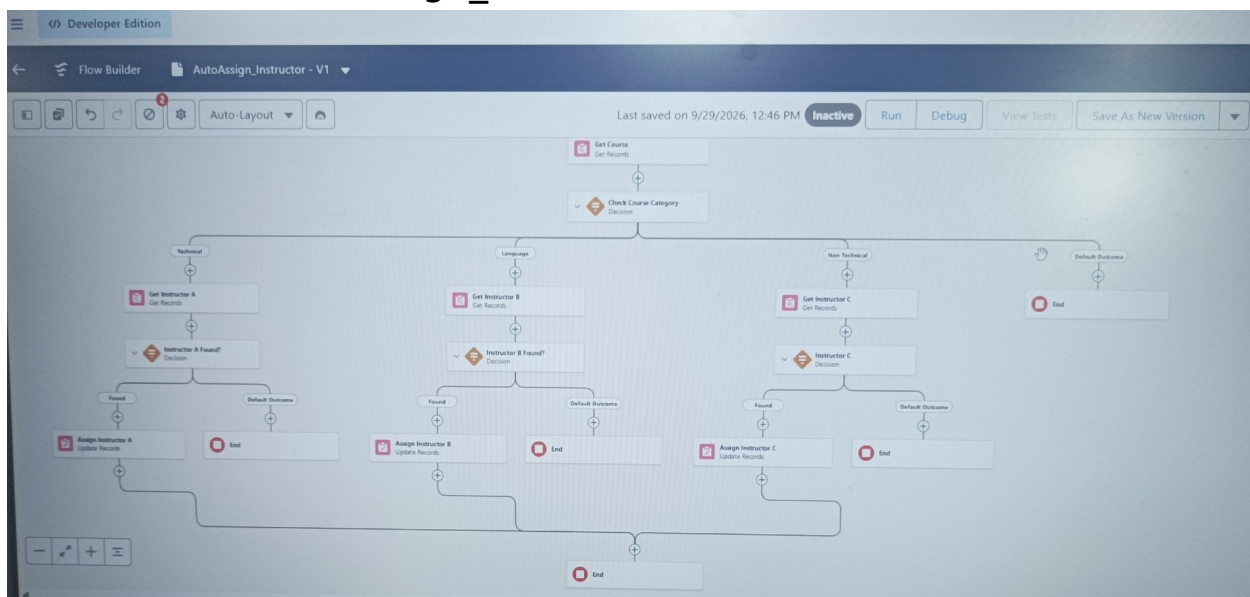
Technical → Instructor A

Language → Instructor B

Non-Technical → Instructor C

Steps:

1. Prepare Instructor__c records with codes (e.g., INSTR_A, INSTR_B, INSTR_C).
2. Create Record-Triggered Flow on Enrollment__c (created/updated).
3. Add Get Records to fetch related Course__c.
4. Add Decision element on Course__r.Category__c:
5. Technical / Language / Non-Technical outcomes.
6. For each outcome:
7. Get Instructor record (A/B/C) based on code or name.
8. Update Records: set Instructor__c = Get_Instructor_X.Id.
9. Save as AutoAssign_Instructor → Activate.



Activity 4: Auto-Create Task for Follow-Up

Purpose:

When an Enrollment is in the "Requested" status, a follow-up task is created for the Admin.

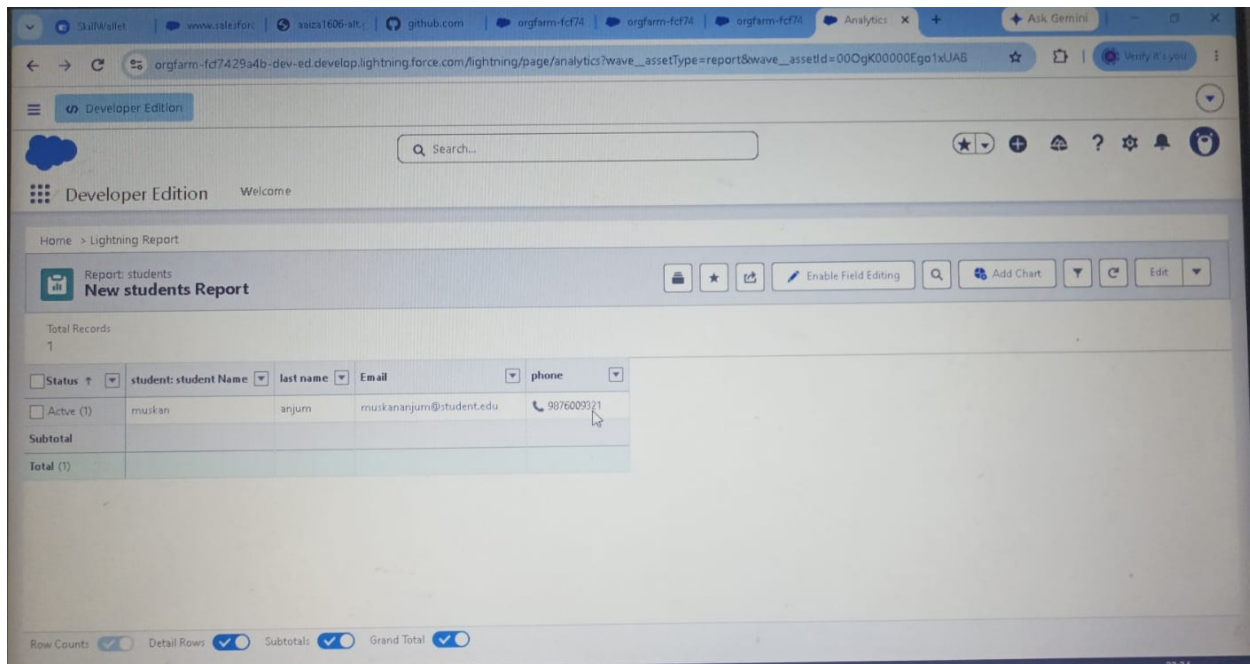
Steps:

1. Create Record-Triggered Flow on Enrollment__c.
2. Trigger: Record created or updated; condition Enrollment_Status__c = "Requested".
3. Add Get Records to fetch Admin user (by Username / Profile).
4. Add Formula (Date): TODAY() + 2 for due date.
5. Add Create Records for Task:
6. Subject: Follow up on enrollment request.
7. OwnerId: Admin user Id.
8. WhatId: Enrollment Id.
9. ActivityDate: formula Today + 2.
10. Save as Create_Followup_Task → Activate.

Milestone 11 – Reports:

Activity 1: Create “Students by Status” Report

1. Go to Reports → New Report.
2. Report Type: Students.
3. Group rows by Status__c.
4. Add columns as needed (Name, Email, Phone).
5. Save as Students by Status.



Activity 2: Create “Enrollments by Course” Report

- 1.New Report → Report Type: Enrollments.
- 2.Group rows by Course__r.Name.
- 3.Show Record Count per course.
- 4.Save as Enrollments by Course.

Activity 3: Create “Pending Enrollment Approvals” Report

- 1.New Report → **Enrollments** report type.
- 2.Filter: **Enrollment_Status__c = "Requested"**.
- 3.Add columns (Student, Course, Owner, Created Date).
- 4.Save as **Pending Enrollment Approvals**.

Activity 4: Create “Revenue Report (Fees Paid)” Report

- 1.New Report → Enrollments report type.
- 2.Filter: Fees_Paid__c = TRUE.
- 3.Group by Course__r.Name.
- 4.Summarize Course_Fee__c or a similar fee field.
- 5.Save as Revenue Report – Fees Paid.

Milestone 12 – Dashboards:

Activity 1: Create “Student Management Dashboard”

Steps:

- 1.Go to **Dashboards** → **New Dashboard**.
- 2.Name: **Student Management Dashboard**.
- 3.Select folder → **Create**.
- 4.Select the new Dashboard folder and Save.

Add components:

- **Donut Chart** → using **Students by Status** report.
- **Bar Chart** → using **Enrollments by Course** report.
- **Gauge** → using **Revenue Report – Fees Paid** (sum of revenue field).
- **Table** → using **Pending Enrollment Approvals** report.

- 4.Arrange components on canvas.

Click **Save** → **Done**.

Milestone 13 – Security Setup

Activity 1: Configure Profiles

1. Go to Setup → Profiles → New (or clone an existing profile). Profiles required:
 - **Admin** – Full access.
 - **Enrollment Officer** – Create/Update Enrollments.
 - **Instructor** – Read-only access to assigned Enrollments.
2. **Steps (example for Enrollment Officer): Profile Name: Enrollment Officer.**
3. **In Object Settings:**
 - **Enrollment__c**: Read, Create, Edit.
 - **Student__c** / **Course__c**: Read (and Edit if required).
4. Save.

Repeat the same steps for the Instructor profile with Read-Only access on Enrollments.

Activity 2: Create Permission Sets

1. Go to Setup → Permission Sets → New.
2. Create **Custom Object Access** (additional access to project objects if needed).
3. Create **Dashboard Editor** (Create/Customize Reports & Dashboards permissions).

4. Assign to relevant users.

Activity 3: Configure Sharing & OWD

Steps:

1. Go to **Setup → Sharing Settings**.
2. Set **Organization-Wide Defaults**:
 - **Students (Student__c)**: Public Read/Write.
 - **Courses (Course__c)**: Public Read/Write.
 - **Enrollments (Enrollment__c)**: Private.
3. Save.

For Instructor access to assigned Enrollments, optionally:

- Add a **User lookup** on Enrollment (Instructor User).
- Use a simple **Record-Triggered Flow** to set **OwnerId = Instructor User** so they can see only their own records.

Milestone 14 – Conclusion

The Salesforce-based Student Enrollment & Course Management System streamlines the complete student academic lifecycle, from student registration and course creation to enrollment management, instructor assignment, approvals, revenue tracking and reporting. By using Salesforce automation features such as validation rules, flows, approval processes and dashboards, we improved data accuracy, operational efficiency and decision-making.

The project shows how a manual academic process can be turned into a scalable, reliable and user-friendly digital solution.